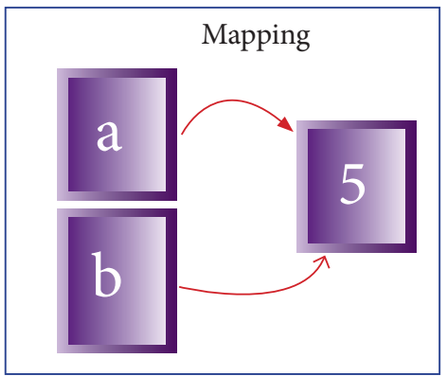
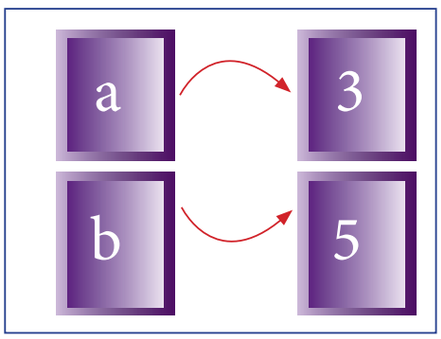


Now consider the following extension of the above statements

1. a:=5
2. b:=a
3. a:=3

If you then change ***a*** to be equal to ***3***, a budding programmer might expect ***b*** also be equal to 3, but that is not the case. ***b*** is still mapped (or pointing) to the integer value of 5. The only thing that changed is ***a***, which is now mapped to the integer value 3.

Mapping after changing the value of ***a***



The scope of a variable is that part of the code where it is visible. Actually, to refer to it, you don't need to use any prefixes then. Let's take an example,

1. **Disp**():
2. a:=7

When you try to display the value of ***a*** outside the procedure the program flags the error ***"name 'a' is not defined"***. This is because the lifetime of the variable is only till the end of the procedure. Also, ***the duration for which a variable is alive is called its 'life time'.***

3.3 LEGB rule

Scope also defines the order in which variables have to be mapped to the object in order to obtain the value. Let us take a simple example as shown below:

1. x:= 'outer x variable'
2. **display():**
3. x:= 'inner x variable'
4. print x
5. display()

When the above statements are executed the statement (4) and (5) display the result as

Output

outer x variable

inner x variable

Above statements give different outputs because the same variable name ***x*** resides in different scopes, one inside the function display() and the other in the upper level. The value ***'outer xvariable'*** is printed when ***x*** is referenced outside the function definition. Whereas when display() gets executed, ***'inner x variable'*** is printed which is the ***x*** value inside the function definition. From the above example, we can guess that there is a rule followed, in order to decide from which scope a variable has to be picked.

The ***LEGB*** rule is used to decide the order in which the scopes are to be searched for scope resolution. The scopes are listed below in terms of hierarchy (highest to lowest).

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